

Automated Diagnosis of Dental Conditions in Panoramic X-Rays with Ensemble Deep Learning Models

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ABSTRACT

Over 85% of clinical dental treatment decisions are based on radiographic interpretation, recent research has shown. This holds particularly for conditions that cannot be easily diagnosed during a routine check-up. Focusing on common pathologies in dentistry (such as caries and impacted teeth) and restorative materials (such as fillings and implants), this research recommends applying deep learning to automatically interpret panoramic dental X-ray images. Correct treatment planning and long-term patient management rely upon accurate detection of such conditions.

Manual interpreting of radiographs is time consuming and prone to human errors. Some of the advanced deep learning methods our approach employs to address this challenge and deliver fast, reliable, and precise image analysis include Convolutional Neural Networks (CNNs), transfer learning, and ensemble modelling. We enhance classification performance and stability by adopting ensemble methods that tap into the strengths of multiple deep models.

The system, being more than 92% accurate in its classification, had a potential for advancing oral healthcare using AI, enhancing clinical outcomes, and minimizing diagnostic delays.

Index Terms—Deep learning, Dental Radiography, Disease Detection, Restoration, Ensemble Model, CNN, Transfer learning

I. INTRODUCTION

Artificial intelligence in dental radiographic analysis: a new paradigm Analysis has resulted in tremendous progress in diagnostic precision and working efficiency. Panoramic radiography continues to be a keystone in the diagnosis of oral pathology extensive visualization of teeth and facilitating

detection of both the disease and restoration conditions in line. Deep learning technology has changed dental practice diagnostics and patient treatments substantially during recent years. Panoramic dental radiography serves as an important diagnostic tool for clinical dental practice according to ADA statistics [1]. Clinical procedures require diagnostic radiographic analysis according to ADA standards and these examinations reveal hidden issues which affect treatment decisions in more than 80% of cases, while dental procedures depend on radiographic analysis in 90% of situations. The demand for swift, accurate diagnostic solutions has led deep learning to become an emerging automated system that improves radiograph reading accuracy. Multi-layered neural networks in deep learning duplicate cognitive decision systems to acquire significant relationships from intricate imaging data [2], [3]. Compared to other dental challenges, the restricted availability of annotated radiographic datasets remains a permanent issue because of patient privacy rules and time-consuming expert annotation procedures [2]. This situation hinders deep learning models from becoming effective in various dental populations and imaging scenarios. The current setting suggests that transfer learning, together with ensemble modeling, provides essential benefits. The pre-trained transfer learning approach helps models learn from extensive image data collections and then adjust to new details from dental tasks, even when presented with scarce datasets. Simultaneously, ensemble modeling employs multiple models to strengthen performance while decreasing prediction uncertainty. Medical imaging professionals have witnessed significant success from these techniques, which now gain interest for dental diagnostic tasks including caries detection and bone loss assessment, and restoration classification [4], [5].

The level of agreement among dental professionals for detecting early-stage caries or restoration defects reaches just 40–60% based on studies [6]. The need for dependable automated systems becomes clear because they help dentists

detect crucial findings with higher consistency and pace. The efficiency of practitioner patient care, along with better clinical results, stems from reducing diagnostic subjectivity.

The research explores deep learning model development for detecting two prevalent dental diseases along with two dental restoration types from panoramic X-ray pictures. The proposed method combines four transfer learning models ResNet-50 [7], InceptionV3 [8] and DenseNet121 [9] and VGG19 [10] with a personalized Convolutional Neural Network (CNN) and a total of four ensemble methods comprising Weighted Average model [11], Stacking model [12], Boosting model [13], and Bagging model [14] to enhance classification accuracy. A high diagnostic accuracy of 93% emerged from the ensemble deep learning framework, which showed its effectiveness in improving dental radiology clinical assessments.

II. RELATED WORK

Disease diagnosis from medical images has been revolutionized by deep learning, effectively identifying patterns in radiographic data, complex in nature. One challenge, however, has been the limitation of annotated data sets, specifically for new diseases such as COVID-19. Jadon [15] introduced few-shot deep learning through transfer learning and data augmentation to generalize using fewer data points. From a few chest X-rays, the model was able to identify COVID-19 anomalies effectively. Pretraining networks, along with synthetic dataset augmentation, mitigated dependency on large annotated data sets, evidencing potential in data-scarce diagnostic applications of a few-shot learning approach.

Deep ensemble learning has also emerged as effective in increasing medical image classification accuracy simultaneously. Visuña and Mendoza [16] presented a computer-aided diagnosis system employing a voting ensemble of six CNNs—VGG16, VGG19, InceptionV3, ResNet101V2, DenseNet121, and CheXNet—to classify chest X-rays (COVID-19, Tuberculosis, Viral Pneumonia, Healthy). Voting ensemble applied model consensus, solving differences by voting using the highest performing model; stacking ensemble employed a three-neural layer network incorporating dropout and ReLU activation to learn non-linear pattern predictions. Both performed significantly better in classification, where stacking proved to have higher accuracy along with robust stability, showcasing the advantages of ensemble methods in reducing overfitting, stabilizing heterogeneous medical image data classification, and providing higher accuracy results.

Likewise, deep learning has proven effective in segmentation and classification in dental radiography tasks. Ghahremani et al. [17] presented a CNN pipeline for automatic detection and segmentation of dentitions and restorations from panoramic dental radiographs. They integrated preprocessing pipelines, region-based CNNs, and morphological processing, together enhancing detection and segmentation accuracy. Diversity of the dataset, reflecting realistic imaging environments, increased generalizability. The system was able to determine segmentation accuracy above 94% and high overlap (Dice coefficient). Grad-CAM visualizations offered

transparency, increasing clinical acceptance and trust, proving explainable AI effective in assisting clinicians in precisely identifying dental anomalies.

Although these works together endorse deep learning success in radiology, a number of issues remain outstanding. Scarcity of large-scale annotated data still stands as a pressing problem, particularly for domain-specific tasks such as dental anomaly detection. Adaptability across domains, patient populations, and variations of imaging equipment is also an ongoing issue. Clinical uptake needs interpretability, where transparent visualizations are needed to portray model decisions. Combining a few-shot learning [15], ensemble learning [16], and explainable segmentation [17] together has the potential to move robust, reliable AI-aided radiology diagnosis forward.

III. METHODOLOGY

A. Dataset Description

The data was downloaded from Roboflow, which hosting labeled medical images. It consists of 1,075 panoramic dental X-rays stored as JPEG, having associated JSON files with class labels and bounding box coordinates for all teeth. It has four labels for dental restoration and states: *caries* (class 0), *filling* (class 1), *impacted teeth* (class 2), and *implants* (class 3). Panoramic images had multiple independently labeled teeth. Therefore, preprocessing was done to identify teeth separately for easier classification. Individual teeth were cropped from panoramic radiographs using a Python script with OpenCV, using the coordinates of the annotations. Cropped images were saved in PNG format, categorized by class labels into respective directories. Following preprocessing, the dataset was expanded from 1,075 panoramic images to 1,814 cropped tooth images, systematically split into a training, validation, and test subsets. Subset distribution for each tooth class is given below in Table I. Additionally, Figure 1 is a data sample.

TABLE I: Detailed Distribution of the Dataset across Training, Validation, and Test Sets

Dataset Split	Total Images	Caries (0)	Filling (1)	Impacted (2)	Implant (3)
Training	1,357	163	608	186	400
Validation	273	36	122	30	85
Testing	184	18	80	24	62

B. Preprocessing and Augmentation

Initial panoramic X-rays from Roboflow had to be heavily preprocessed for effective model training. Having cropped teeth using OpenCV, the image of each tooth was normalized to the same input size. Images were scaled to 224×224 pixels, matching the input requirements of our convolutional neural networks (CNNs). Only InceptionV3 was an exception, which performed better empirically at 229×229 pixels due to its architecture and intrinsic spatial resolution demands [8]. Given the modest size of the dataset, overfitting was a pressing concern. This was addressed by using data augmentation to artificially augment the training set, improving the generalization of the models. Random flipping along the horizontal,

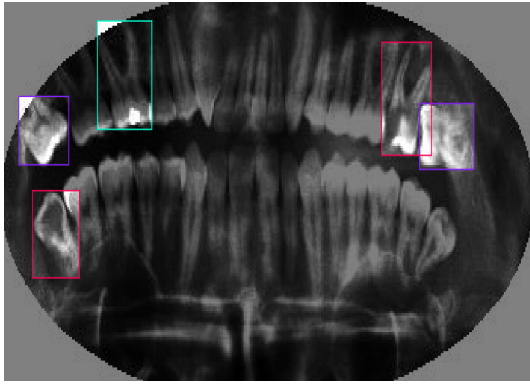


Fig. 1: Panoramic X-Ray Image, Red: Cavity, Blue: Impacted, Green: Fillings

minor rotations ($\pm 10^\circ$), minor zoom ($\pm 10\%$), and brightness changes simulating real-world fluctuations were applied. All these techniques were successful at augmenting data diversity, enhancing model generalization to validation as well as test sets.

Initial model performance was largely determined by image contrast and resolution. Conventional image improvement methods, such as histogram equalization and CLAHE, were tried first. While effective for a wide range of medical applications, these methods did not produce substantial improvement in our dental radiographs. Finer features that separated restorations from anomalies were left insufficiently improved, leading to inaccurate classification. Therefore, we applied Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN) [18], using generative adversarial networks to upscale images with a significant boost in clarity and details. ESRGAN greatly enhanced the visual clarity of cropped teeth images, boosting the test accuracy of CNN by around 3%. This success attested to the efficacy of ESRGAN in addressing visual ambiguities left unsolved by conventional methods. Accordingly, the preprocessing and data augmentation played a crucial role in dataset optimization, sharpening of the images, as well as maintaining model performance consistency in later training and testing stages.

C. Model Architectures and Hyperparameter Tuning

By the complexity associated with dental radiography, choosing appropriate deep learning architectures with optimal hyperparameters was a prerequisite for effective feature extraction as well as classification. We employed custom-created as well as existing architectures: a custom Convolutional Neural Network (CNN), ResNet-50 [7], InceptionV3 [8], DenseNet121 [9], as well as VGG19 [10].

Initial attempts at reapplying successful configurations were suboptimal, highlighting systematic hyperparameter tuning unique to our data. Keras Tuner with Bayesian Optimization [19] was employed, efficiently querying the hyperparameter search space from prior trials with vastly decreased computation compared to random search or grid search. The primary hyperparameters tuned include dropout rates (0.2–0.4),

number of dense layers (1–5), number of units for a dense layer (16–512, increments of 16), and RMSprop learning rates $\{1 \times 10^{-2}, 1 \times 10^{-3}, 1 \times 10^{-4}\}$.

The overall hyperparameter optimization process included the following steps:

- 1) Hyperparameter search space definition.
- 2) At most 100 trials of Bayesian search for each architecture, repeated twice for validation.
- 3) Saving the model only if there's a validation accuracy improvement.
- 4) Stopping early after five epochs if there's no improvement in validation accuracy for the sake of computational efficiency.

The optimization process provided model-specific architectures with a substantial performance improvement. Below are described the optimized architectures.

a) **Custom CNN**: CNNs are best at identifying subtle visual attributes automatically, which is perfect for dental X-ray examination. Our model of CNN consists of five convolutional blocks with MaxPooling, filter sizes of 144, 192, 176, 208, 192, with Flatten and a dropout rate of (0.3), and three dense blocks of 144, 192, 176 neurons.

b) **ResNet-50**: ResNet-50 utilizes residual links to avoid vanishing gradients, helping learning of deep representations [7]. Transfer learning was utilized by freezing pre-trained convolutional units and augmenting a tailored dense architecture (416, 512, 512, 512, 16 neurons) with a dropout rate of (0.2).

c) **InceptionV3**: It utilizes multiple-scale visual information through parallel convolutions of varying dimensions, helpful for fine dental radiograph patterns [8]. Image inputs were scaled to 299×299 pixels. Transfer learning involved frozen convolutional layers and a single dense layer (512 units) with dropout (0.2).

d) **DenseNet121**: DenseNet121 facilitates gradient flow and feature reuse through dense connections, enhancing detailed image classification [9]. Transfer learning involved frozen convolutional layers followed by a dense architecture (480, 16, 16, 16, 16 neurons) and dropout (0.3).

e) **VGG19**: Simple but effective with small data, VGG19 employs tiny (3×3) convolution kernels [10]. Transfer learning with frozen pretrained convolution layer, single dense layer (512 units) with dropout rate of (0.3).

f) **Models Structure Summary and Comparison of Performance**: All of the models employed a four-neuron softmax output layer (caries, filling, impacted, implant). Following optimization, the most accurate was the custom CNN, with ResNet-50 and VGG19 having a very similar performance. DenseNet121 and InceptionV3 were a little behind. CNN's high performance underscored its applicability for dental radiographic nuances, with ResNet-50 and VGG19 having strong generalization, validating ensemble methods.

1) **Ensemble Methods for Performance Improvement**: To yield a more accurate prediction and take advantage of complementary strengths of the optimized models, we examined ensemble techniques: Weighted Average [11], Stacking [12], Bagging [14], and Boosting [13]. These approaches combined

individual predictions, reducing variance, enhancing generalizability, and stabilizing classification.

a) **Weighted Average Ensemble:** Weighted average ensemble method was employed, incorporating softmax probability predictions of the top three models (custom CNN, ResNet-50, and VGG19), with optimized weights from validation accuracies (0.45 for CNN, 0.25 for ResNet-50, 0.30 for VGG19). The weights were adjusted to minimize the variance of the predictions, yielding better classification performance compared to each model separately.

b) **Stacking Ensemble:** Stacking ensemble with stacked generalization [12] first combined all optimized models' predictions. Following empirical examination, redundant models as well as noise were removed, leaving CNN, ResNet-50, VGG19, and the Weighted Average Ensemble. Deep-level features were obtained with layer slicing to bypass Sequential API constraints, subsequently normalized using scikit-learn's StandardScaler. XGBoost was substituted for Random Forest as the meta-learner because of its better performance with high-dimensionality as well as non-linear interactions. XGBoost's hyperparameter setting (moderate tree depth, reduced learning rate) considerably increased accuracy as well as class-wise recall.

c) **Bagging Ensemble:** Bagging (Bootstrap Aggregating) combines independently learned base models over several bootstrapped training subsets. Majority voting aggregation [14] greatly minimized variance, enhancing test set generalization. This significantly minimized data noise and outlier sensitivities prevalent for medical imaging tasks.

d) **Boosting Ensemble:** Gradient Boosting progressively updates model predictions with a focus on previously misclassified cases, enhancing harder cases progressively [13]. Boosting was always beneficial, with a strong improvement for harder dental X-ray classifications.

e) **Summary and Performance Ranking:** All models had a common output structure of four classes (caries, filling, impacted, implant). Experimental performance placed Weighted Average as the top performer with the highest accuracy, with Boosting as a strong second, followed by Bagging. Stacking initially had little improvement, but having removed redundant models, normalized features, and implemented XGBoost as a meta-learner, greatly enhanced reliability and interpretability, as a complementary approach to our pipeline.

IV. EXPERIMENTS AND RESULTS

A. Experiment 1: Performance of Individually Tuned Models

The evaluation of deep learning model performance for dental issue classification from panoramic X-ray images occurred through Keras Tuner experiments. Our objective focused on identifying top architecture implementations among our custom CNN design and four established transfer learning models that included ResNet50, InceptionV3, DenseNet121, and VGG19. The key parameters for dropout rates as well as dense layers' number and size along with learning rates, underwent adjustments through Bayesian Optimization. The

model received between 100 and 200 tuning rounds, conducted twice for each round to maintain result consistency. We applied the optimal configuration selection from validation accuracy optimization to conduct the final training procedure before testing. We assessed the models by testing them on 184 tooth images, which belonged to four categories which included caries, filling, impacted, and implant. The evaluation of performance included an assessment of accuracy and precision, together with recall through macro average calculations to ensure class balance across different class sizes. Table II summarizes the evaluation results for each individually tuned model.

TABLE II: Performance of Individually Tuned Models (Experiment 1)

Model	Accuracy (%)	Precision (%)	Recall (%)
Custom CNN	88.59%	88.52%	88.04%
ResNet-50	85.87%	86.26%	85.33%
InceptionV3	61.41%	62.09%	61.41%
DenseNet121	65.76%	76.39%	29.89%
VGG19	82.07%	86.67%	77.72%

The custom CNN delivered superior overall results by demonstrating a slight advantage over ResNet-50 and VGG19 as per the data in Table II. The recall performance of DenseNet121 was deficient compared to its precision output since it struggled to correctly identify most medical conditions particularly for cases with scarce samples. InceptionV3 had a lower performance rate than other models when evaluated because its deeper structure proved ineffective for generalization for picking models for use in ensemble strategies.

B. Experiment 2: Evaluation of Ensemble Models

To provide a comparative summary of each ensemble model's performance, Table III provides a comprehensive comparison of ensemble models across multiple evaluation metrics, including accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and AUC-ROC.

TABLE III: Performance Comparison of Ensemble Methods

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
Weighted Average	94.00	92.93	92.93	0.92
ViT-SL	90.70	91.00	91.10	0.91
Siamese Network	86.24	86.09	84.24	0.84

We conducted an analysis to assess the impact of Weighted Average, Stacking, Boosting, and Bagging techniques on the total classification results using the base model predictions or deep features from CNN, ResNet-50, and VGG19. The ensemble models integrated the best base models' (CNN and ResNet-50 and VGG19) predictive outputs or deep features into combined prediction systems, which underwent the same evaluation metrics as Experiment 1.

a) **Weighted Average Ensemble:** The ensemble model processed three model predictions through softmax while combining them with optimized weights, which distributed 0.45 to CNN, 0.25 to ResNet-50, and 0.30 to VGG19. Among all ensemble types, this method produced the highest performance

through reaching an accuracy and precision, and recall rates of **92.93%**. The figure illustrating its confusion matrix in Figure 2 indicates excellent and uniform class identification performance.

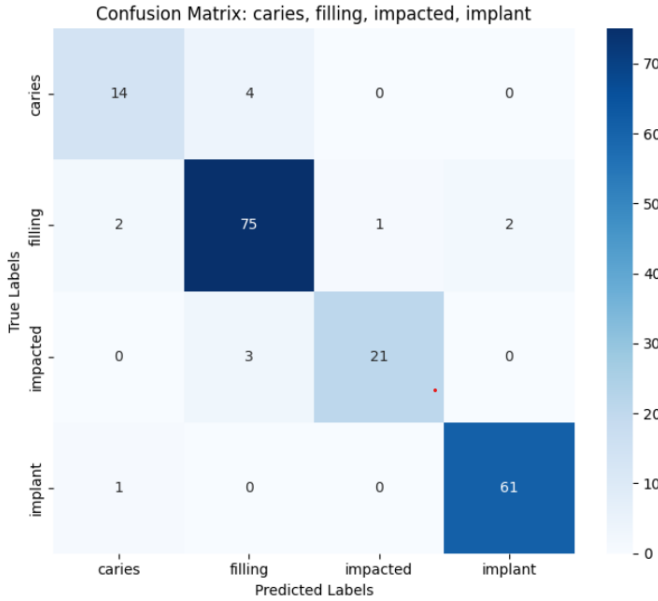


Fig. 2: Confusion Matrix for Weighted Average Ensemble

b) Stacking Ensemble: Multiple successive improvements were implemented for the stacking ensemble. The ensemble started with all base models before being minimized to contain only CNN and ResNet-50, together with VGG19 and the Weighted Average Ensemble due to multiple noise factors and model redundancy. The model extracted vectors from sliced layers between the CNN architectures by removing classification layers, which were then normalized through StandardScaler to standardize numerical values. The use of XGBoost as our meta-learner emerged because it works efficiently with large input dimensions and effectively handles nonlinear relationships between features [13]. The improved system configuration generated **88.59%** accuracy together with **89.02%** precision and **88.60%** recall.

c) Boosting Ensemble: The ensemble system utilized Gradient Boosting, which performed sequential improvement by adapting each iteration to fix errors detected in the previous model predictions. The experimental results showed remarkable performance because the system achieved **90.76%** accuracy together with **90.87%** precision and **90.57%** recall metrics.

d) Bagging Ensemble: Numerous base models tackled different training subsets from the data through the bagging ensemble approach. The straightforward implementation resulted in good performance metrics with an accuracy, precision, and recall reaching **89.13%**.

e) Conclusion of Experiment 2: The experimental results in Experiment 2 confirm that ensemble techniques enhance classification accuracy for dental anomaly detection.

Among the ensemble techniques, Weighted Average Ensemble achieved the highest success rate, with Boosting in second position and Bagging and Stacking ranking next. The stacking ensemble reached superior performance through model reduction, together with feature standardization and XGBoost integration, as these enhancements establish vital demands for medical imaging system success.

f) real tested data: We tested our models on real data of four images, two for impacted, one for filling, and one for cavity and all were classified correctly

V. DISCUSSION

The research evaluated state-of-the-art deep learning technology for the diagnosis of dental anomalies (caries, fillings, impacted teeth, implants) from panoramic X-rays. Two main experiments compared standalone deep learning models with ensemble methods with constrained data. An architecture-tailored CNN attained a rate of 88.59% accuracy, outperforming ResNet-50, InceptionV3, DenseNet121, and VGG19. Results affirm that architecture-specific designs provide superior dental radiograph assessment results. Performance of InceptionV3 and DenseNet121 degraded because of challenges while handling changes in data distribution as well as image scales; performance of InceptionV3 was only 61.41% despite having shown success in common sight tasks. Experiment 2 indicated that ensemble techniques greatly increased model robustness as well as generalization. Weighted Average ensemble (CNN, ResNet-50, VGG19) yielded the highest precision/recall rates as well as accuracy (92.93%). Model fusion greatly addressed individual model deficiencies. Boosting as well as Bagging ensembles increased reliability even more, while the optimized Stacking ensemble, with reduced-size models as well as XGBoost as meta-learner, proved to have strong adaptive ability. The research indicated that specialist architectures alongside ensembles enhanced diagnosis with limited data. Image clarity was enhanced by pre-processing with the aid of ESRGAN, resulting in a 3% accuracy improvement, highlighting the significance of quality inputs for medical imaging. However, several limitations emerged:

a) Limited dataset diversity and size: Limited dataset diversity and size restricted generalizability to other populations or imaging types.

b) Overfitting Risk: Regularization and data augmentation did not entirely mitigate overfitting with high-capacity models, necessitating more diverse data sets or sophisticated domain adaptation techniques.

c) Computational Resource Constraint: High computational needs of the ensemble methods of boosting and stacking may present a problem for real-time clinical applications under resource-limited environments.

d) Lack of Interpretation Tools: Absence of explainable AI (Grad-CAM, SHAP) implementation hindered model interpretation, discouraging clinical adoption.

e) Lack of Clinical Testing: Models were not validated with clinical trials, nor with external benchmarks to provide practical clinical utility.

Panorama diagnosis is greatly enhanced with ensemble-integrated deep learning models with normalization. Low-data settings can be aided even with a few-shot learning with ensemble assistance [15], [16].

VI. CONCLUSION AND FUTURE WORK

Utilizing ensemble deep models in panoramic X-ray images, this study provides a comprehensive framework for automatic dental condition diagnosis. Based on our comprehensive research, we made a series of key findings: First, the proposed customized CNN architecture outperformed benchmark transfer learning models (ResNet-50, VGG19, etc.) with an accuracy of **88.59% accuracy** when using it independently. Second, our Weighted Average ensemble approach, which integrates CNN, ResNet-50, and VGG19, proved ensemble methods applicable for dental radiograph interpretation through achieving state-of-the-art performance metrics for **92.93% accuracy, 92.93% precision**, and **92.93% recall**. Third, our preprocessing pipeline based on ESRGAN super-resolution, which significantly **3% accuracy improvement**, verified that image enhancement is essential for medical image applications.

Three factors primarily explain our accomplishment: (1) well-designed architecture specification and hyperparameter tuning through consideration of dental radiography's nature; (2) ensembling complementary models through intelligent hybridization; and (3) high-performance image preprocessing for discriminative feature enrichment. Consistent with our goal to make AI-assisted diagnosis available for clinical deployment, our findings have practical significance for clinical practice since they show that AI-assisted diagnosis can be improved to expert performance levels while reducing time-to-interpret and variability.

Limitations and Future Directions

While our results are promising, several limitations point to valuable directions for future research:

- **Data diversity and availability:** We trained our models using a small dataset (1,814 tooth images cropped from full-face images). Future studies will need to utilize larger datasets having variability in demographic populations (e.g., ethnicity, socioeconomic level) and in devices used for imaging, environmental conditions for imaging (e.g., clinical diagnosis, research), and in clinical settings (e.g., academic, private practice) so that results would be generalizable. A collaboration with a dental institution would allow overcoming data shortages while maintaining patient privacy.
- **Computational efficiency:** The ensembles, even if accurate, will be costly in terms of computational resources. Developing light versions of ensemble methods via model pruning, knowledge distillation, or neuroarchitecture search might enable real-time deployment in clinically constrained computing situations.
- **Clinical interpretability:** There are no inherent explanation capabilities in the current system. Implementation of

multiple visualisation techniques, such as Grad-CAM or attention mechanisms, would enable clinicians to comprehend what the model is deciding and create confidence and collaboration between clinicians and AI along the diagnosis pipeline.

- **Comprehensive clinical validation:** Whereas our technical validation yields some encouraging evidence, no doubt exists that clinical trials are necessary to determine real world performance. Future research must determine how the system integrates into clinical workflows, how clinicians actually use the system in practice, and lastly whether and how it affects relevant patient outcome.
- **Expanded diagnostic scope:** The current system takes into account four conditions for consideration (caries, fillings, impacted teeth, and implants). If we could include periodontal diseases in this count, adding the consideration of assessment of root canals to it, and even early diagnosis of oral cancers, we could maximize clinical utility.

We see three promising research areas in the future: firstly, data scarcity problems can be solved with the help of few-shot learning techniques, given that they are able to learn from a limited amount of examples; secondly, multi-task techniques for detection, segmentation and classification, could increase the extent to which dentists are supported diagnostically; and finally, research and development for standard benchmarks for dental (and comparable) AI systems will allow for a fair comparison and for more progress to be made in AI systems for dentistry.

These advances, together with ongoing collaboration between researchers in AI and dentists, will be an important force in effective translation from our technical achievements to practical tools for enhancing oral care worldwide. The scheme presented in this report is a good foundation for any advances in dental diagnostics using AI.

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